

INTRODUCTION

The two world wars, from 1914 to 1918, and from 1939 to 1945, were the most destructive conflicts in human history. In terms of technology, the period was marked by the spectacular development of aviation and of motorized warfare, especially tanks.

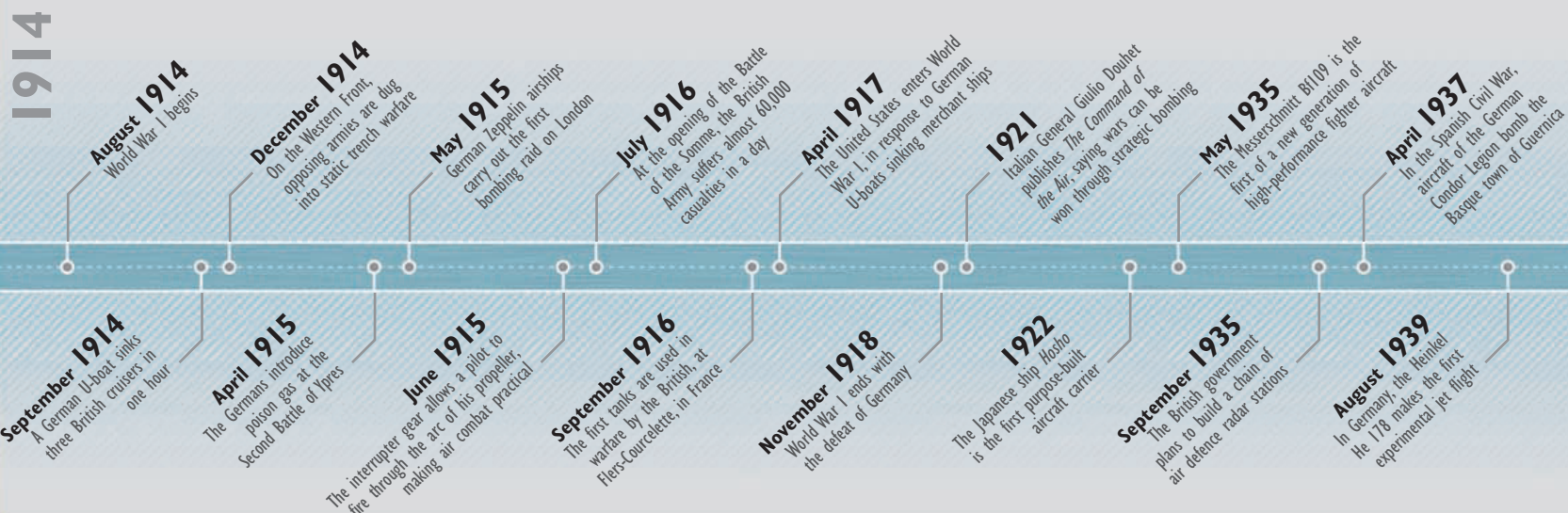
The pace of development was by no means even. Infantry firepower increased over time, from the introduction of light machine-guns and mortars, to sub-machine-guns, automatic rifles and, late in World War II, assault rifles. But there is no comparison between this scale of change and the absolute transformation of aerial warfare. Armies entered World War I with a few hundred flimsy, unarmed reconnaissance aircraft. By 1945, however, fleets of thousands of multi-engined bombers were able to devastate enemy cities, while ground forces could barely operate if their opponents controlled the skies. Dominance in naval battles shifted from large battleships

to aircraft carriers, while submarines established themselves, unexpectedly, as potentially war-winning commerce raiders.

The tank, a useful – if clumsy – adjunct to World War I infantry operations, was vital to the fast-moving motorized warfare of World War II, when coordinated by radio with ground-attack aircraft serving as aerial artillery. The motor truck eventually made horse-drawn transport obsolete, although, as late as 1941, Germany still used hundreds of thousands of horses for its invasion of the Soviet Union.

The pressure of war generated a search for “wonder weapons”, as scientists were enlisted to help the military effort. Poison gas proved indecisive in World War I, and chemical weapons were not used between 1939 and 1945. The jet aircraft, V-1 flying bomb, and V-2 ballistic missile were all introduced by Germany late in World War II, without affecting the outcome of the conflict; however, the atom bomb, developed by the United States between 1942 and 1945 at vast cost, truly heralded a new era in warfare.

KEY DATES



THE BATTLE OF VERDUN – 1916



THE BATTLE OF ARRAS – 1917



US GROUND FORCES – c.1943

